

A Machine Learning Framework for Toxicity Prediction in the Tox21 Dataset Using Hybrid Molecular Representations

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Abstract—Aim: This study aims to develop and evaluate a machine learning pipeline for binary toxicity classification, focusing on the NR-AhR (Aryl Hydrocarbon Receptor) pathway using the Tox21 dataset.

Methods: Several classifiers, including Logistic Regression, Support Vector Machines (SVM), and Random Forest, were trained on different molecular representations: 2D molecular descriptors, Morgan fingerprints, and a combined feature set. The pipeline included data pre-processing, feature selection (Mutual Information and Recursive Feature Elimination), hyperparameter optimization via GridSearchCV, and model evaluation on a held-out test set.

Results: Combining physicochemical descriptors with structural fingerprints consistently improved predictive performance. The best-performing model, a Support Vector Machine trained on mixed features, achieved a Precision-Recall AUC of 0.723, along with the highest F1-score (0.667) and MCC (0.622). Random Forest achieved the highest ROC-AUC (0.926), although this did not translate into superior performance on threshold-dependent metrics.

Conclusion: The proposed pipeline demonstrates that integrating complementary molecular representations enhances toxicity prediction. The results also highlight the importance of selecting appropriate evaluation metrics for imbalanced datasets, as PR-AUC and MCC provide a more informative assessment than ROC-AUC in this setting.

Index Terms—Toxicity prediction, Tox21, Machine Learning, Molecular Descriptors, Morgan Fingerprints, Cheminformatics, NR-AhR.

1 INTRODUCTION

THE assessment of chemical toxicity is a fundamental pillar of public health and environmental protection. With thousands of new compounds synthesized annually, traditional *in vivo* toxicological testing—which is ethically sensitive, costly, and time-consuming—is no longer sufficient to meet regulatory demands. The Toxicology in the 21st Century (Tox21) initiative was launched to address this gap by leveraging high-throughput screening (HTS) to evaluate the biological activity of small molecules across a wide range of pathways.

1.1 Biological Background: The AhR Pathway

The Aryl Hydrocarbon Receptor (AhR) is a ligand-activated transcription factor that plays a crucial role in the detection

of and adaptation to chemical exposure. It is particularly known for mediating the toxicity of environmental pollutants such as dioxins. Activation of the NR-AhR pathway can lead to significant biological changes, including altered xenobiotic metabolism, immunotoxicity, and developmental defects. Predicting AhR activation is therefore a key objective in modern toxicological modeling.

1.2 Machine Learning in Cheminformatics

Machine learning (ML) has revolutionized the field of Quantitative Structure-Activity Relationship (QSAR) modeling. By encoding chemical structures into numerical vectors—descriptors and fingerprints—ML algorithms can learn complex mappings between chemical structure and biological activity. In particular, supervised learning allows for the creation of predictive models that can rapidly screen vast virtual libraries of compounds, prioritizing those with favorable safety profiles.

1.3 Study Objectives

This work presents a study project aimed at designing and evaluating a machine learning pipeline for the binary classification of compounds as active or inactive toward the NR-AhR target. Although not intended as a production-grade or clinical study, the project follows a rigorous experimental design to emulate a realistic cheminformatics workflow.

Specifically, this work aims to:

- 1) Assess the impact of different molecular representations (2D descriptors, Morgan fingerprints, and their combination) on predictive performance.
- 2) Investigate feature selection strategies to reduce redundancy and improve model stability in high-dimensional cheminformatics data.
- 3) Compare the performance of linear and non-linear classifiers under a consistent evaluation framework.
- 4) Properly address class imbalance inherent to high-throughput screening (HTS) data using class weighting and appropriate evaluation metrics (e.g., PR-AUC).

2 MATERIALS

The dataset employed in this study is derived from the Tox21 initiative, specifically focusing on the NR-AhR (Aryl Hydrocarbon Receptor) bioassay. This dataset consists of chemical compounds annotated based on their ability to activate the AhR signaling pathway, providing a binary classification task (active vs. inactive).

The initial dataset included 7,831 compounds. After pre-processing—removing molecules that could not be successfully parsed and discarding samples with missing labels for the selected endpoint—a final dataset of 6,542 compounds was obtained. This filtering step ensures consistent supervision during model training, at the cost of reducing the available data.

2.1 Feature Description

Chemical information was represented using two complementary feature types, capturing both global physicochemical properties and local structural patterns:

- **Molecular Descriptors (2D):** A set of 217 numerical features computed using RDKit, describing global properties of the molecules. These include physicochemical attributes (e.g., molecular weight, logP), topological descriptors, and surface-related measures such as Topological Polar Surface Area (TPSA). These features provide an interpretable representation of molecular characteristics relevant to biological activity.
- **Morgan Fingerprints:** Circular fingerprints generated with radius 2 and a dimensionality of 2048 bits. Each bit encodes the presence or absence of specific substructures derived from local atomic environments. Unlike descriptors, these features capture fine-grained structural information and are particularly effective in modeling structure–activity relationships.

In addition to using each representation independently, a combined feature set was constructed by concatenating molecular descriptors and Morgan fingerprints. This hybrid representation aims to leverage both global and local information to improve predictive performance.

2.2 Software and Libraries

All experiments were conducted in Python. Data manipulation and preprocessing were performed using `pandas` and `NumPy`. Cheminformatics operations, including molecule parsing, sanitization, and feature extraction, were carried out using the `RDKit` library.

Machine learning tasks—including feature scaling, feature selection, model training, and evaluation—were implemented using `scikit-learn`. In particular, utilities such as `StandardScaler`, `SelectKBest`, `RFE`, and `GridSearchCV` were employed to construct a consistent and reproducible pipeline.

Data visualization, including exploratory plots and performance curves (e.g., Precision-Recall and ROC curves), was performed using `matplotlib` and `seaborn`.

3 METHODS

3.1 Data Preparation and Cleaning

Raw SMILES strings were converted into RDKit molecule objects. Molecules that failed sanitization or could not be successfully parsed were discarded. In addition, samples with missing labels for the selected endpoint (NR-AhR) were removed to ensure consistent supervision.

The final dataset consists of 6,542 compounds, with 768 active (11.7%) and 5,774 inactive (88.3%) samples. This pronounced class imbalance is a critical aspect of the problem and directly influences both model training and evaluation strategy.

3.2 Exploratory Data Analysis (EDA)

Exploratory data analysis was performed to characterize the dataset, assess feature behavior, and identify potential issues such as class imbalance and feature redundancy.

3.2.1 Dataset Characterization and Target Selection

A preliminary exploratory analysis was conducted to assess the distribution of labels and the extent of missing values across all toxicity targets. For each endpoint, we computed the number of non-toxic (0), toxic (1), and missing entries, along with their corresponding proportions. This step is critical, as both class imbalance and missing data can significantly affect model performance and generalization.

Figure 1 shows the distribution of samples for each target, highlighting the imbalance between toxic and non-toxic classes as well as the presence of missing values.

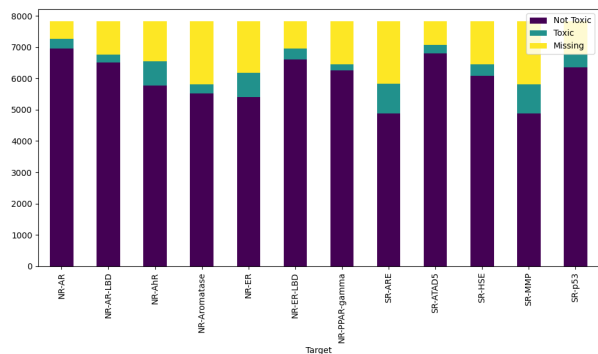


Fig. 1: Class distribution across all targets, including non-toxic, toxic, and missing samples.

The analysis reveals two main challenges. First, most targets exhibit a strong class imbalance, with toxic samples representing only a small fraction of the available labeled data. Second, several endpoints contain a substantial proportion of missing values, reducing the effective dataset size available for supervised learning.

The selection of the target variable therefore requires a trade-off between data quantity and class balance. Choosing a target with too many missing values would significantly reduce the number of usable samples, while selecting a highly imbalanced target would make the learning task more difficult and potentially bias the model toward the majority class.

Among the available endpoints, NR-AhR was selected as a reasonable compromise. Although it contains a non-negligible proportion of missing values (approximately 16%), it still retains a sufficient number of valid samples for training. At the same time, its proportion of positive (toxic) instances (around 11.7%) is relatively higher compared to other targets, which often exhibit more extreme imbalance.

It is important to note that this choice does not eliminate the challenges associated with imbalance and missing data, but rather mitigates them to a manageable extent. Alternative strategies, such as multi-task learning across multiple targets, could potentially exploit additional information, but were not considered in this work in order to maintain a simpler and more controlled experimental setting.

3.2.2 Feature Distributions

To assess the discriminative potential of individual descriptors, features were ranked using Mutual Information (MI), and the top 10 were visualized using violin plots (Figure 2). Several features exhibit clear differences in distribution between active and inactive compounds, suggesting their relevance for classification.

It is important to note that this analysis is purely exploratory: since it is performed on the full dataset, it is not used to drive feature selection in order to avoid data leakage.

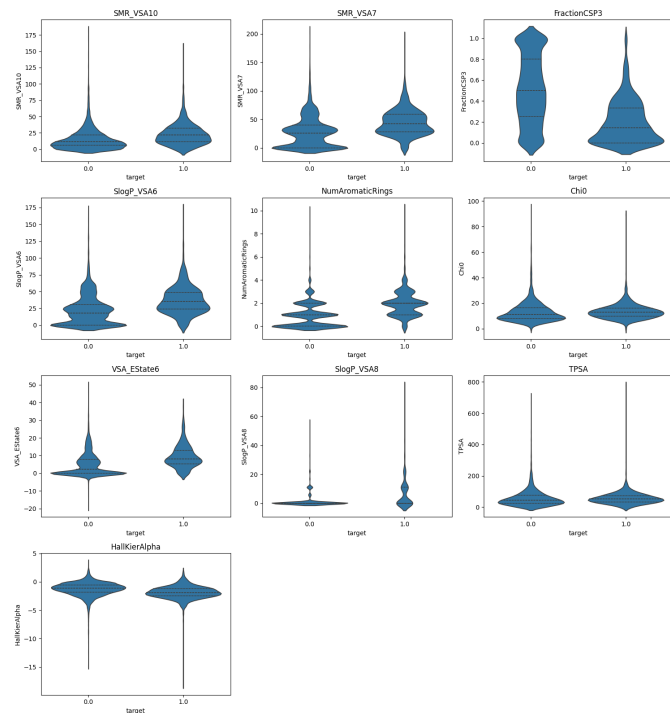


Fig. 2: Violin plots of the top 10 descriptors ranked by Mutual Information. Distinct distribution patterns between classes are observed for several features.

3.2.3 Correlation Analysis

To quantify linear relationships between descriptors, Pearson’s correlation coefficient was computed. The coefficient r is defined as:

$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}} \quad (1)$$

Due to the high dimensionality of the descriptor space, instead of visualizing the full correlation matrix, we focused on identifying highly correlated feature pairs. The analysis revealed 94 pairs of descriptors with strong correlation ($r > 0.95$), indicating substantial redundancy in the feature space and motivating the application of feature selection techniques.

3.3 Feature Engineering and Scaling

All preprocessing steps were performed exclusively on the training data to prevent information leakage. Continuous features were standardized to have zero mean and unit variance:

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

where μ and σ are estimated on the training set and subsequently applied to the test set. This transformation is essential for scale-sensitive models such as Support Vector Machines and Logistic Regression, ensuring stable optimization and comparable feature contributions.

3.4 Feature Selection Strategy

Given the high dimensionality and the presence of strong multicollinearity among descriptors, a two-stage feature selection strategy was applied. Feature selection was performed using only the training data to avoid optimistic bias.

- 1) **Univariate Selection:** Features were ranked using Mutual Information (MI), which captures both linear and non-linear dependencies between each feature X and the target Y :

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) \quad (3)$$

This step provides a model-agnostic ranking of individual feature relevance.

- 2) **Recursive Feature Elimination (RFE):** A Logistic Regression model with regularization was used to iteratively remove less informative features based on coefficient magnitude. This step accounts for feature interactions within a multivariate setting.

The final feature set was defined as the intersection of features selected by both methods, retaining only those consistently identified as informative. This conservative strategy prioritizes robustness, resulting in 129 descriptors.

3.5 Model Training and Evaluation

Three supervised learning algorithms were selected to capture different modeling assumptions and levels of complexity:

- **Logistic Regression (LR):** A regularized linear classifier serving as a strong and interpretable baseline.

- **Support Vector Machine (SVM):** A margin-based classifier evaluated with multiple kernels (linear, radial basis function, and polynomial) to account for both linear and non-linear decision boundaries.
- **Random Forest (RF):** An ensemble method based on decision trees, capable of modeling non-linear relationships and feature interactions while being relatively robust to noise.

Hyperparameter tuning was performed using grid search with 5-fold stratified cross-validation on the training set. Stratification preserves class proportions across folds, mitigating bias introduced by class imbalance.

Model selection was based primarily on the area under the Precision-Recall curve (PR-AUC), which provides a more informative evaluation than accuracy or ROC-AUC in imbalanced datasets by emphasizing performance on the positive class. ROC-AUC was reported as a complementary, threshold-independent metric.

To further assess classification performance under class imbalance, two threshold-dependent metrics were computed: the Matthews Correlation Coefficient (MCC) and the F1-score. MCC provides a balanced measure that accounts for all entries of the confusion matrix, while the F1-score emphasizes the trade-off between precision and recall for the positive class.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (4)$$

Final model performance was evaluated on a held-out test set that was not involved in either training or hyperparameter optimization, ensuring an unbiased estimate of generalization performance.

4 RESULTS

4.1 Overall Model Performance

The overall performance of the evaluated models across different feature representations is summarized in Table 1. Results are reported in terms of PR-AUC, ROC-AUC, F1-score, and MCC, providing both threshold-independent and threshold-dependent perspectives on model performance.

TABLE 1: Comparison of model performance across different feature representations.

Model	Features	PR-AUC	ROC-AUC	F1	MCC
Support Vector Machine	Mixed	0.723	0.911	0.667	0.622
Random Forest	Mixed	0.707	0.926	0.534	0.550
Logistic Regression	Mixed	0.704	0.920	0.624	0.588
Support Vector Machine	Descriptors	0.686	0.898	0.629	0.581
Support Vector Machine	Fingerprints	0.684	0.912	0.633	0.586
Random Forest	Descriptors	0.678	0.916	0.449	0.464
Random Forest	Fingerprints	0.663	0.919	0.500	0.503
Logistic Regression	Fingerprints	0.645	0.907	0.592	0.549
Logistic Regression	Descriptors	0.622	0.910	0.533	0.498

Among the evaluated configurations, models trained on the mixed feature representation consistently achieved the strongest performance in terms of PR-AUC. In particular, the Support Vector Machine with mixed features obtained the highest PR-AUC (0.723), as well as the best F1-score (0.667) and MCC (0.622), indicating a superior balance between precision and recall on the minority class.

Random Forest achieved the highest ROC-AUC (0.926) when trained on mixed features, suggesting strong overall class separability. However, its lower F1-score and MCC indicate reduced effectiveness at the default decision threshold, likely due to the class imbalance.

Logistic Regression showed competitive and stable performance across metrics, particularly in the mixed feature setting, but did not outperform SVM in any of the evaluated criteria.

Across feature representations, Support Vector Machines demonstrated consistently strong performance, with both descriptor-based and fingerprint-based models achieving competitive F1 and MCC scores. This suggests that SVMs are robust to the choice of feature representation.

It is important to note that threshold-dependent metrics (F1 and MCC) were computed using the default decision threshold. No explicit threshold optimization was performed. While this ensures consistency across models, it may not reflect the optimal operating point for each classifier. Future work could explore threshold tuning strategies (e.g., maximizing MCC or F1 on a validation set), particularly in the presence of class imbalance.

4.2 Performance Curves

Figure 3 shows the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for the best-performing model (Support Vector Machine trained on the mixed feature set).

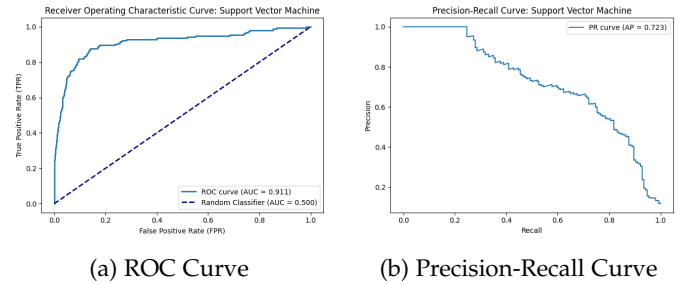


Fig. 3: Performance curves for the Support Vector Machine on the mixed feature representation.

The ROC curve indicates strong overall class separability, consistent with the high ROC-AUC observed for this model. However, in the presence of class imbalance, the Precision-Recall curve provides a more informative assessment. The PR curve highlights the model’s ability to effectively identify the minority (active) class, maintaining a favorable balance between precision and recall across decision thresholds.

4.3 Confusion Matrix Analysis

To further investigate the classification behavior, the confusion matrix of the Support Vector Machine trained on the mixed feature representation was analyzed, as this model provides the best overall performance across multiple metrics.

The confusion matrix revealed 1101 true negatives, 104 true positives, 54 false positives, and 50 false negatives. This

corresponds to a precision of approximately 0.66 and a recall of approximately 0.68 for the positive (toxic) class.

The model exhibits a relatively balanced error profile, with comparable numbers of false positives and false negatives. This balance contributes to stable F1 and MCC scores, indicating a good trade-off between precision and recall.

However, the number of false negatives remains non-negligible, accounting for a substantial fraction of the positive class. In the context of toxicity prediction, this is a critical limitation, as toxic compounds incorrectly classified as non-toxic may pose safety risks.

This behavior reflects the use of a default decision threshold, which does not explicitly optimize the trade-off between precision and recall. In safety-critical applications, a more conservative operating point may be preferable, prioritizing the reduction of false negatives even at the expense of increased false positives. Future work could therefore explore threshold tuning strategies to better align model behavior with domain-specific requirements.

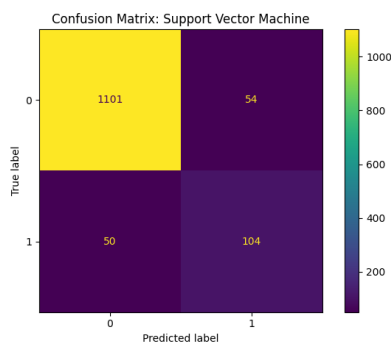


Fig. 4: Confusion matrix for SVM using mixed features evaluated on the held-out test set.

5 DISCUSSION AND CONCLUSIONS

This study presents a machine learning pipeline for toxicity prediction targeting the NR-AhR pathway in the Tox21 dataset. The results show that combining 2D molecular descriptors with Morgan fingerprints yields a richer and more informative representation of chemical space, effectively capturing both global physicochemical properties and local structural motifs.

5.1 Summary of Key Findings

The experimental results support the following observations:

- **Feature Integration:** The combination of molecular descriptors and Morgan fingerprints generally provides the best performance across models, particularly in terms of PR-AUC. This suggests that integrating complementary global and local molecular information is beneficial, although some single-representation models (e.g., SVM with fingerprints) remain competitive.
- **Model Performance:** Support Vector Machines consistently achieved the strongest overall performance,

particularly when trained on mixed features, outperforming both Logistic Regression and Random Forest in terms of PR-AUC, F1-score, and MCC.

- **Evaluation Metrics:** In the presence of class imbalance, PR-AUC and MCC provide more informative performance estimates than ROC-AUC. While several models achieved high ROC-AUC values, these did not always translate into strong performance on threshold-dependent metrics, highlighting the importance of considering multiple evaluation criteria.

5.2 Cheminformatic Interpretability

Beyond predictive performance, several selected descriptors show consistency with known chemical principles underlying AhR activation. In particular, features related to lipophilicity (e.g., $\log P$) and surface-related properties (e.g., SMR_VSA10) are aligned with the known tendency of AhR ligands to be hydrophobic molecules capable of interacting with the receptor binding pocket.

5.3 Limitations and Future Directions

This study is limited by its focus on a single biological endpoint, which restricts the generalizability of the findings across the broader spectrum of Tox21 assays. While the selected NR-AhR endpoint provides a representative and moderately balanced setting, it does not fully capture the heterogeneity observed across all toxicity targets.

Future work will extend the proposed pipeline to the complete set of Tox21 assays and explore more expressive modeling approaches, such as Graph Convolutional Networks (GCNs), which operate directly on molecular graph representations and may reduce the reliance on manually engineered features.

In addition, alternative strategies for handling class imbalance could be investigated, including random undersampling, oversampling techniques, and synthetic data generation methods such as SMOTE, in order to assess their impact on predictive performance under severely skewed class distributions.

Finally, the comparison between models is limited by the absence of formal statistical significance testing. The observed performance differences are relatively small, and may not be statistically distinguishable from random variation. Incorporating statistical testing or repeated cross-validation would provide a more robust assessment of model differences.

5.4 Code Availability

All code used in this study, including data preprocessing, model training, and evaluation pipelines, is publicly available at: <https://github.com/marco-la-rosa/tox21-toxicity-ML-classification>

The repository is structured to facilitate reproducibility and modular experimentation, including environment specifications and documentation.

ACKNOWLEDGEMENTS

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